Interpreting ACF and PACF

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Introduction

The autocorrelation function (ACF) and partial autocorrelation function (PACF) are the foundational diagnostic tools for ARIMA model identification. The Box-Jenkins approach uses their characteristic patterns to guess plausible orders for AR and MA terms before fitting. Even with modern automated tools (`auto.arima`), reading the ACF and PACF remains an essential skill for sanity-checking automated choices, identifying seasonal structure, and diagnosing residuals.

Prerequisites

A working understanding of autoregressive (AR) and moving-average (MA) processes, the difference between correlation and partial correlation, and stationarity as a precondition for meaningful ACF interpretation.

Theory

The ACF ρ_k measures correlation between y_t and y_{t-k}. The PACF ϕ_{kk} measures correlation between y_t and y_{t-k} after removing the effects of shorter lags $1, 2, \ldots, k-1$. Box-Jenkins identification rules:

Process	ACF	PACF
AR(p)	Tails off (decays slowly)	Cuts off at lag p
MA(q)	Cuts off at lag q	Tails off
ARMA(p, q)	Tails off	Tails off

The dashed lines in R's ACF plot show approximate 95 % confidence bands ($\pm 1.96/\sqrt{n}$) under a white-noise null; values outside the bands are statistically significant at the 5 % level.

For seasonal data, expect spikes at multiples of the seasonal period; multiplicative seasonal patterns produce decaying spikes at $s, 2s, 3s, \ldots$ in the ACF.

Assumptions

The series is stationary; if not, difference first until it becomes stationary, then read the ACF / PACF on the differenced series.

R Implementation

```r
library(forecast)

set.seed(2026)
ar_series <- arima.sim(list(ar = 0.7), n = 200)
ma_series <- arima.sim(list(ma = -0.5), n = 200)

par(mfrow = c(2, 2))
acf(ar_series, main = "ACF of AR(1)")
```

```r
pacf(ar_series, main = "PACF of AR(1)")
acf(ma_series, main = "ACF of MA(1)")
pacf(ma_series, main = "PACF of MA(1)")
par(mfrow = c(1, 1))

ggtsdisplay(ar_series)
```

Output & Results

The four-panel display shows characteristic Box-Jenkins patterns: AR(1) ACF tails off geometrically while its PACF cuts off after lag 1; MA(1) ACF cuts off after lag 1 while its PACF tails off. `ggtsdisplay()` integrates the time-series plot with ACF and PACF in a single ggplot-style view.

Interpretation

A reporting sentence: "The ACF of the simulated AR(1) tailed off geometrically (lag-1 correlation 0.69, decaying smoothly); the PACF cut off sharply after lag 1 (0.69 at lag 1, well within bands at lags 2 onward), consistent with the AR(1) data-generating process. The MA(1) showed the reverse pattern, confirming Box-Jenkins identification rules." Identify the model class from the patterns before fitting any model.

Practical Tips

- Always difference the series until stationary before reading ACF / PACF on the levels; non-stationary series produce decaying ACFs that mimic AR.
- Use `forecast::ggtsdisplay()` for a combined time-series + ACF + PACF view; faster than three separate plots.
- Significance bands assume a white-noise null; after model fitting, the residual ACF is the key diagnostic and should be approximately within bands.
- Seasonal patterns appear at lags equal to the seasonal period; spikes at $s, 2s, 3s$ on the ACF indicate seasonal AR/MA structure.
- Complex ACF / PACF patterns (mixed ARMA, seasonal-plus-non-seasonal) are difficult to identify by eye; reach for `auto.arima` and verify the chosen orders against the ACF.
- For very long series, even tiny correlations are statistically significant; focus on the magnitude and pattern, not on which lags fall just outside the bands.

R Packages Used

Base R `acf()` and `pacf()`; `forecast::ggtsdisplay()` for ggplot-style integrated diagnostics; `feasts::ACF()`, `PACF()`, `gg_tsdisplay()` for the modern tidyverts equivalent; `astsa::acf2()` for compact two-panel display.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Time series basics